# Deep Matting for Cloud Detection in Remote Sensing Images

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***Abstract*— Cloud detection, as an important preprocessing operation for remote sensing (RS) image analysis, has received increasing attention in recent years. Most of the previous cloud detection methods consider the detection as a pixel-wise image classification problem (cloud versus background), which inevitably leads to a category-ambiguity when dealing with the detection of thin clouds. In this article, starting from the RS imaging mechanism on cloud images, we re-examine the cloud detection under a totally different point of view, i.e., to formulate cloud detection as a mixed energy separation between foreground and background images. This process can be further equivalently implemented under a deep learning-based image matting framework with a clear physical significance. More importantly, the proposed method is capable to deal with three different but related tasks, i.e., "cloud detection," "cloud removal," and "cloud cover assessment," under a unified framework. The experimental results on the three satellite image data sets demonstrate the effectiveness of our method, especially for those hard but common examples in RS images, such as the thin and wispy cloud.**

***Index Terms*— Cloud detection, convolutional neural networks (CNNs), deep learning, image matting, remote sensing (RS) image.**

## I. Introduction

THE rapid development of remote sensing (RS) technology in recent years has opened a door for people to better understand the Earth. Optical RS, as a large family of RS imaging technologies, has been extensively applied in many areas in recent years, such as land monitoring, disaster relief, an military reconnaissance. Despite its wide applications, the fact that the ground objects in an optical RS image usually being covered by clouds has greatly limited the usage of optical images and increased the difficulty of image analysis. As reported by Stubenrauch *et al.* [1], on average, more than 50% of the Earth surface is covered by clouds every day. The research on cloud detection is of great importance and has received increasing attention in recent years.

Most of the previous cloud detection methods frame the cloud detection as a semantic segmentation process, i.e., to generate a binary mask for the foreground (cloud) and background image regions under a pixelwise classification (cloud versus background) paradigm. Some commonly used methods include the band grouping/thresholding methods [2]–[7], the traditional image segmentation methods [8]–[10], and the recent popular deep learning-based segmentation methods [11]–[17]. As these methods are mostly borrowed from the computer vision community without considering the mechanism behind the RS imaging, the pixelwise classification will inevitably lead to a category-ambiguity in terms of detecting thin clouds. Therefore, a common defect of these methods lies that they are not able to deal with thin clouds properly.

The cloud in an image usually presents in a mixed form of visual appearance with the cloud itself and the ground objects underneath. Cloud may have various thicknesses and present in various transparency. As the energy received by an imaging sensor can be approximated by a linear combination of the reflectance of clouds and ground objects, a RS image can be considered as a superimposition of a "clouds layer" and a "background layer." Therefore, cloud detection is naturally a mixed image separation problem.

In the image processing field, image matting [18]–[20] refers to a group of methods that aim to extract foreground objects from an image, which have very similar idea compared to the above descriptions. Matting is an important task in image and video editing. Some related works can be traced back to the 1990s [18]. Traditional image matting methods can be divided into two large groups of families: 1) sampling-based methods [21]–[23] and 2) propagation-based methods [19], [24], [25], where the former one produces the matte by a predefined metric given a set of foreground and background sampling regions, and the latter one reformulates the prediction as the propagation of foreground and background regions. The matting task usually produces a "matte" that can be used to separate the foreground from the background in a given image, which naturally corresponds to the cloud detection process. To this end, starting from the "mixed energy imaging model (to be introduced)" of cloud images, we propose a brand new cloud detection paradigm called "Deep Cloud Matting," which reformulates the cloud detection as a mixed energy separation between foreground and background images and

Manuscript received January 31, 2020; revised March 16, 2020; accepted April 9, 2020. Date of publication April 30, 2020; date of current version November 24, 2020. This work was supported in part by the National Key Research and Development Program of China under Grant 2017YFC1405605, in part by the National Natural Science Foundation of China under Grant 61671037, in part by the Beijing Natural Science Foundation under Grant 4192034, and in part by the National Defense Science and Technology Innovation Special Zone Project. *(Corresponding authors: Zhengxia Zou; Zhenwei Shi.)*

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Digital Object Identifier 10.1109/TGRS.2020.2988265

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can be equivalently implemented under an image matting framework. As the recent advances in deep learning technology have greatly promoted the progress of image matting [20], [26], we take the advantage of the deep convolutional neural network (CNN) and the multitask learning framework so that the matting-based cloud detection can be implemented by learning to predict multiple outputs, including the "cloud reflectance map" and the "cloud opacity map," under a unified deep convolutional architecture. The proposed framework is scalable and flexible, has clear physical significance, and can be jointly trained in an end-to-end fashion. In particular, we consider the traditional cloud detection model as a special case of our method.

To improve the prediction of some hard examples such as thin and wispy clouds, the attention mechanism is further integrated into our method. Attention was originally introduced in machine translation to improve the performance of an encoder–decoder recurrent neural network (RNN) model by taking into account the input from several time steps to make one prediction [27]. In a CNN-based model, the introduction of mechanism attention helps investigate the spatial correlations of different feature locations, and now, it has been widely used in many computer vision tasks such as object detection [28], optical character recognition [29], and image captioning [30], [31]. In the proposed method, by introducing the cloud foreground map (indicating where there is a cloud) as the pixelwise attention weights to the loss functions of the other two tasks, the learning of the cloud reflectance and opacity can be well-instructed. The abovementioned design has two advantages. First, it makes the training process focuses more on those hard examples. Second, it is helpful to reduce the correlation between the predictions of cloud reflectance and opacity. Our method predicts not only the detailed cloud region but also the accurate cloud reflectance and opacity, which can be further used for cloud removal and cloud cover assessment. The experimental results on the three satellite image data sets have demonstrated the effectiveness of our method.

Contributions of this article are summarized as follows.

1) In most previous cloud detection literature, the detection is framed as a pixelwise classification process without considering the nature of the RS imaging. This pixelwise classification paradigm results in an inherent defect when dealing with the detection of thin and wispy clouds. This article proposes a brand new cloud detection framework, which reformulates cloud detection as a foreground-background energy separation process. This idea can be further implemented under a classical image matting framework, which is derived from a mixed energy imaging model of cloud images.
2) To improve the detection of some "hard examples" such as the thin and wispy cloud, the attention mechanism is introduced in our method to reduce coupling between tasks and make the learning process focuses more on those hard examples.
3) In previous RS literature, cloud detection [2]–[17], cloud cover assessment [32]–[34], and cloud removal [35]–[42] are investigated separately, despite the high correlation between them. This inhibits joint optimization and makes the implementation of the methods highly complicated. Instead of designing individual algorithms, the proposed method deals with the three tasks under the same framework and is trained in an end-to-end fashion.

The rest of this article is organized as follows. In Section II, we introduce the mixed energy imaging model of cloud images and how cloud detection is formulated under an image matting framework. In Section III, we give a detailed introduction to our proposed method, including network configuration, multitask loss function, and implementation details. In Section IV, we introduce the data set used in our experiments. Some experimental results are given in Section V. In Section VI, we discuss the drawbacks and limitations of our method, and the conclusions are drawn in Section VII.

## II. Mixed Energy Imaging Model of Cloud Image

When a satellite or an aircraft flies over a cloud-covered area, the onboard imaging sensors receive the reflectance energy of the ground objects and the cloud at the same time. The amount of energy received by the sensors per unit of time can be approximately considered as a linear combination of the three terms [43], [44]: 1) the reflectance energy of the clouds; 2) the reflectance energy of the ground objects without the interference of the clouds; and 3) the radiation of ground objects, as shown in Fig. 1. This process can be described as follows:

$$\begin{aligned} E &= E_{\mathrm{CR}} + (1-\alpha)(E_{\mathrm{BR}} + E_{\mathrm{BE}}) \\ &\approx E_{\mathrm{CR}} + (1-\alpha)E_{\mathrm{BR}} \end{aligned} \tag{1}$$

where $E$ represents the total energy received by the sensor, $E_{\mathrm{CR}}$ represents the energy reflected by the clouds, $E_{\mathrm{BR}}$ represents the reflectance energy of the ground objects without the occlusion of clouds, and $E_{\mathrm{BE}}$ represents the radiation of ground objects, which can be usually neglected for visible bands. $\alpha$ is an attenuation factor of the ground reflectance due to the occlusion of clouds ($0 \leq \alpha \leq 1$). The larger the $\alpha$, the thicker the cloud will be: $\alpha = 0$ means there is no cloud while $\alpha = 1$ means the ground object is completely occluded by the cloud. We refer to the abovementioned model as the "Mixed Energy Imaging Model" of cloud images.

To this end, a RS image $I(x)$ can be generally expressed as a linear combination of a cloud reflectance map $R_c(x)$ and a background reflectance map $R_b(x)$

$$I(x) = R_c(x) + [1-\alpha(x)]R_b(x) \tag{2}$$

where $x$ represents the pixel locations in an image.

According to the abovementioned model, we deal with three different problems, i.e., cloud detection, cloud cover assessment, and cloud removal, under a unified framework.

### A. Task 1: Cloud Detection

As the cloud reflectance map $R_c(x)$ defines how much energy is reflected by clouds per unit time, the cloud detection

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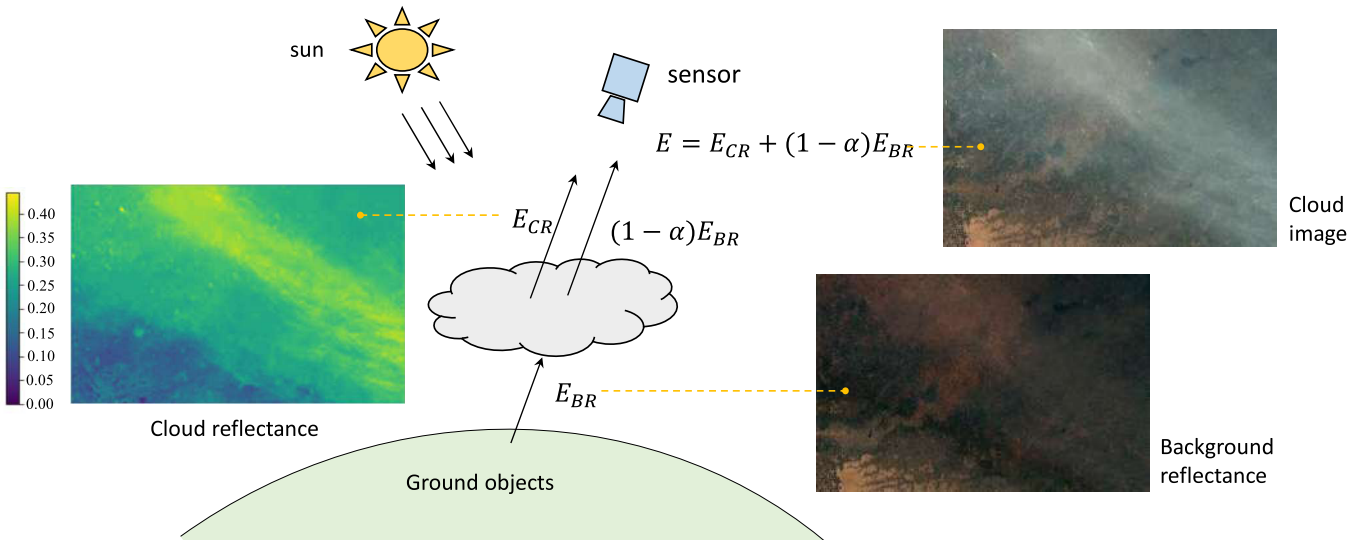



Fig. 1. Illustration of the "Mixed Energy Imaging Model" of cloud images [43], [44]. The energy received by a sensor per unit time can be approximately considered as a linear combination of the reflectance energy of the cloud $E_{\mathrm{CR}}$ and the ground objects $E_{\mathrm{BR}}$.

task can be, thus, considered as the learning of a mapping from the input cloud image to the cloud reflectance map: $I(x) \mapsto R_c(x)$. When $\alpha(x) = 1$ and $R_c(x)$ is set to binary values (i.e., 0 and 1), the prediction will degenerate to a traditional cloud detection method, which simply ignores the reflectance.

### B. Task 2: Cloud Cover Assessment

As $\alpha(x)$ corresponds to how much of the reflectance energy of ground objects has been suppressed due to the occlusion of clouds, we define it as the "thickness" of the cloud. Therefore, the cloud cover assessment tasks can be considered as the learning of a mapping from the input to the cloud opacity map: $I(x) \mapsto \alpha(x)$.

### C. Task 3: Cloud Removal

Cloud removal is essentially a background image recovery problem. According to (2), it is easy to obtain the background reflectance image as follows:

$$R_b(x) = \frac{I(x) - R_c(x)}{1 - \alpha(x)}, \quad \alpha(x) \neq 1. \tag{3}$$

This means once we have obtained $R_c(x)$ and $\alpha(x)$, the cloud can be easily removed and background images can be, thus, recovered. It should be noticed that when $\alpha(x) = 1$, the ground is completely covered by clouds and, thus, cannot be recovered.

## III. Deep Cloud Matting

In recent years, deep CNN has played a central role in many computer vision tasks, such as image classification [45], [46] and object detection [47]–[49]. CNN has also been extensively applied in a variety of RS tasks, such as object detection [50], scene labeling [51], RS image captioning [52], and cloud detection [12]. A CNN model learns high-level image representations with better discrimination and robustness by constructing multiple layers of neural networks as opposed to those in traditional methods, where image features are designed manually. In this article, we build our network based on a CNN architecture.

### A. Networks Architecture

We formulate the learning and prediction of the cloud reflectance map $R_c(x)$ and the cloud opacity map $\alpha(x)$ under a multitask learning framework. The networks consist of an encoder and multiple decoders, where the encoder aims to learn high-level feature representations of the input image and the decoders aim to predict multiple desired outputs throughout multiple heads, as shown in Fig. 2.

To improve the prediction of hard examples, e.g., thin cloud, we further integrate the attention mechanism into our model by introducing an additional attention branch to the decoders. Therefore, the decoder consists of three output branches: the first one aims to predict cloud reflectance map $R_c(x)$, the second one aims to predict the cloud opacity map $\alpha(x)$, and the last one, i.e., the attention branch, aims to generate binary cloud mask for cloud pixels, meanwhile, to instruct the learning of the previous two branches and help them concentrate more on difficult regions. Specifically, the attention branch takes in the foreground mask $A(x)$ (where 1 is the cloud-covered pixels and 0 is the no cloud pixels) as its ground truth reference. The predicted attention scores are employed as pixelwise weights of the loss functions of the other two tasks (to be introduced in Section III-B).

As a CNN model consists of a series of convolutional and pooling layers, features in deeper layers will have stronger invariance but less equivariance. Although this could be beneficial to category recognition, it usually suffers the loss of details such as the object's edge and boundary. To improve the learning of features with both high-level semantics and local details, the feature fusion (FF) is employed in our method by

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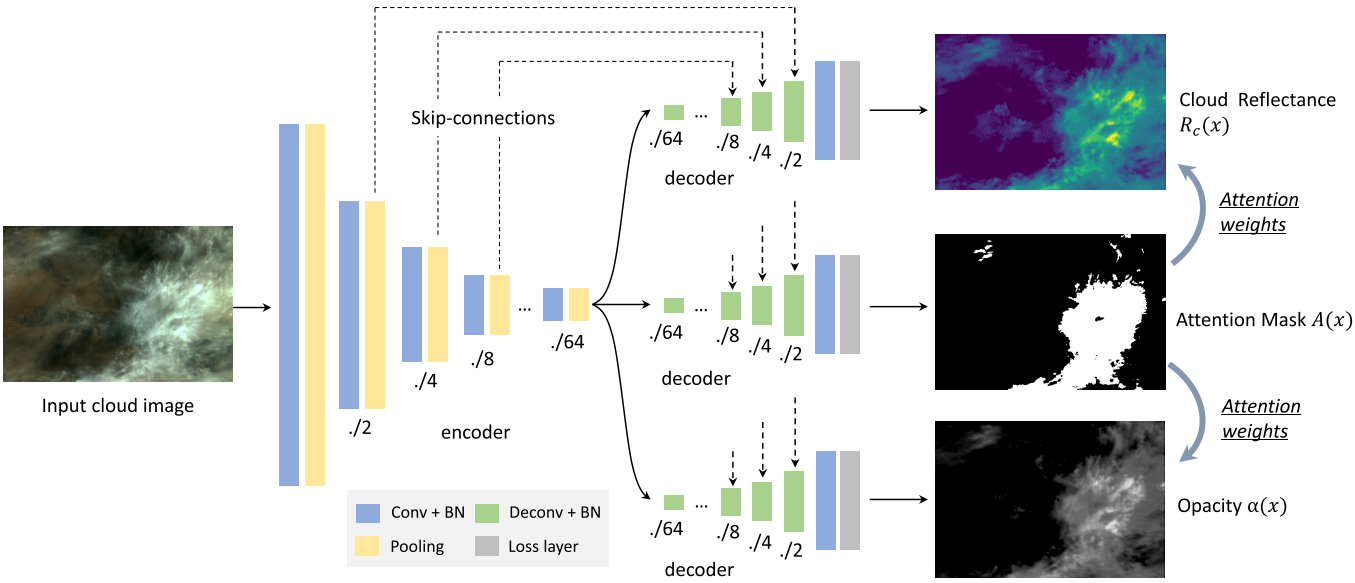



Fig. 2. Overview of the architecture of the proposed networks. The networks consist of an encoder for learning feature representation and three decoders for predicting multiple outputs including cloud reflectance, cloud opacity, and the cloud mask (attention map).

introducing the skip-connection across different layers from the encoder to the three decoders, as shown in Fig. 2. The abovementioned networks can be end-to-end trained with the help of a multitask loss function.

### *B. Multitask Loss Function*

Our loss function $L$ consists of three parts: 1) the loss of attention branch $L(A(x))$; 2) the loss of cloud reflectance prediction $L(R_c(x))$; and 3) the loss of cloud opacity prediction $L(\alpha(x))$

$$L = \sum_x (\gamma_1 L(A(x)) + \gamma_2 L(R_c(x)) + \gamma_3 L(\alpha(x))) \quad (4)$$

where $\gamma_1$, $\gamma_2$, and $\gamma_3$ are three positive coefficients for balancing the learning weights between three tasks. $x$ is the pixel locations. The details of the three loss functions are described as follows.

*1) Attention Branch:* The prediction of the attention map is essentially a pixelwise binary classification process. We use the binary cross-entropy loss as its loss function

$$L(A(x)) = -\widehat{A}(x)\log(A(x)) - (1 - \widehat{A}(x))\log(1 - A(x)) \quad (5)$$

where $A(x)$ and $\widehat{A}(x)$ are the predictions (probabilities) and binary labels of the attention branch.

*2) Cloud Reflectance Branch:* As the reflectance of a cloud pixel is a continuous value, we formulate the prediction of cloud reflectance as a regression problem. To obtain more robust prediction, especially for the outliers, e.g., some thin clouds with low reflectance, we use the $L1$ (absolute value) function as the loss of this branch

$$L(R_c(x)) = A(x)|R_c(x) - \widehat{R_c}(x)| \quad (6)$$

where $R_c(x)$ and $\widehat{R_c}(x)$ represent the predicted and true cloud reflectance values, respectively. $A(x)$ is used as pixelwise weights of the loss function to make the learning focuses on cloud regions.

TABLE I
DETAILED CONFIGURATION OF OUR NETWORKS

| | Layer | Input | Ker | Stride | #Ker | $\sigma(\cdot)$ |
|---|---|---|---|---|---|---|
| **Encoder** | conv_pool1 | image | 3x3 | 2 | 64 | ReLu |
| | conv_pool2 | conv_pool1 | 3x3 | 2 | 128 | ReLu |
| | conv_pool3 | conv_pool2 | 3x3 | 2 | 256 | ReLu |
| | conv_pool4 | conv_pool3 | 3x3 | 2 | 256 | ReLu |
| | conv_pool5 | conv_pool4 | 3x3 | 2 | 512 | ReLu |
| | conv_pool6 | conv_pool5 | 3x3 | 2 | 512 | ReLu |
| | conv7 | pool6 | 3x3 | 1 | 512 | None |
| **Decoder**: 1~3 | deconv1 | conv7 | 3x3 | 2 | 512 | ReLu |
| | deconv2 | deconv1 + conv6 | 3x3 | 2 | 512 | ReLu |
| | deconv3 | deconv2 + conv5 | 3x3 | 2 | 256 | ReLu |
| | deconv4 | deconv3 + conv4 | 3x3 | 2 | 128 | ReLu |
| | deconv5 | deconv4 + conv3 | 3x3 | 2 | 128 | ReLu |
| | deconv6 | deconv5 + conv2 | 3x3 | 2 | 64 | ReLu |
| | conv_output | deconv6 + conv1 | 3x3 | 1 | 1 | None |

*3) Cloud Opacity Branch:* Cloud opacity can also be learned through a regression process. The loss function is defined as follows:

$$L(\alpha(x)) = A(x)|\alpha(x) - \widehat{\alpha}(x)| \quad (7)$$

where $\alpha(x)$ and $\widehat{\alpha}(x)$ represent the predicted and true cloud opacity values. $A(x)$ is used as pixelwise weights of the loss function to make the learning focuses on cloud regions.

### *C. Implementation Details*

*1) Default Setting:* We build a seven-layer convolutional network as our encoder and another seven-layer convolutional network as our decoder. The detailed configuration of our networks is shown in Table I. The columns "Ker," "Stride," "#Ker," and "$\sigma(\cdot)$" mean the size of the convolutional kernel,

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TABLE II
SUMMARY OF OUR GF-1 EXPERIMENTAL DATA SET

| | | |
|---|---|---|
| Image Info. | # total imgs | 328 |
| | img size (pxl) | 4,500×4,500, 12,000×13,400 |
| | resolution | 8m/pxl, 16m/pxl |
| | source | GaoFen-1 PMS and WFV |
| training set | # imgs | 72 |
| | # thin cloud imgs | 0 |
| | # thick cloud imgs | 38 |
| | # no cloud imgs | 34 |
| testing set | # imgs | 256 |
| | # thin cloud imgs | 99 |
| | # thick cloud imgs | 134 |
| | # no cloud imgs | 23 |

the convolutional or pooling stride, the number of filters, and the kind of the nonlinear activation layers, respectively. The rows "conv" and "deconv" mean the convolution operation and the deconvolution operation, respectively, [53] (a.k.a. the transposed convolution), which is used for upsampling the feature map. Apart from the output layer, batch normalization (BN) [54] is embedded in all convolution and deconvolution layers to speed up training. The decoders of the three tasks have the same architecture. Since the output map of our attention branch does not require high accuracy, a rough guide is enough for training. Therefore, we set $\gamma_1 = 1$ and $\gamma_2 = \gamma_3 = 10$. We used the Adam optimizer [55] with a learning rate of $10^{-4}$ for training. We train at a batch size 3 for 16 epochs.

*2) Data Augmentation:* In order to increase the diversity of training data and reduce the gap between real data and synthetic data, extensive data augmentation methods are adopted. For an image of a size $512 \times 512$, we first rotate it with the angle randomly selected from [0°, 90°, 180°, 270°]. Then, a patch with a size of $410 \times 410$ is randomly cropped from the rotated image and resized to $512 \times 512$. Finally, we randomly flip the augmented image.

## IV. EXPERIMENTAL DATA SET

Our experimental data set consists of 328 RS images that are captured by the GaoFen-1 (GF-1) satellite. The original images have two resolutions: 8 m/pixel from the panchromatic and multispectral (PMS) sensor with the image size of about $4500 \times 4500$ pixels, and 16 m/pixel from the wide field-of-view (WFV) sensors with the image size of about $12000 \times 13000$ pixels. The statistics of our data set are given in Table II. There are 72 images in our training set and 256 images in our testing set. Each image has been downsampled to a fixed size, $512 \times 512$ pixels, for training. Since the raw pixel of the original GF-1 data has four bands (blue, green, red, and infrared) and is 16-bit depth, all images are converted to 8-bit RGB images before they are fed into the networks. Apart from that, we do not perform any other preprocessing operations. The data set covers the most types of ground features, such as city, ocean, plains, plateaus, glacier, desert, and gobi. Each image in our data set has been manually

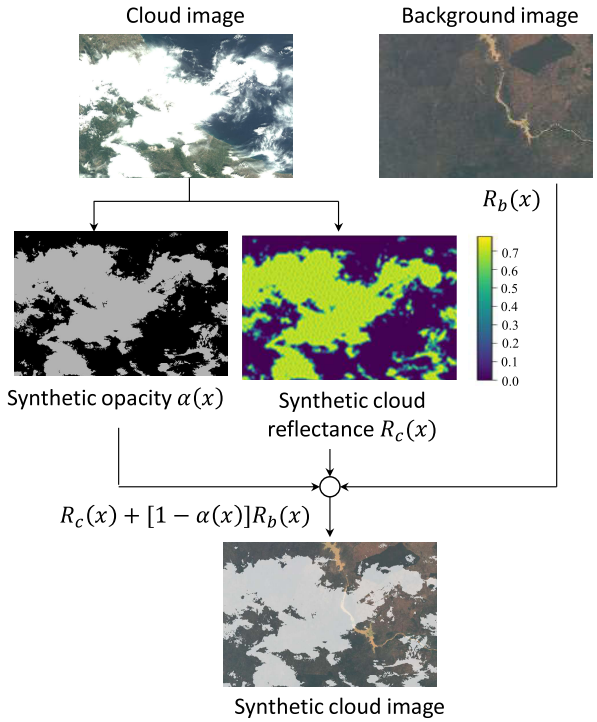



Fig. 3. Illustration of the synthetic training data generation process.

labeled with a pixelwise binary cloud mask as the ground truth of the attention branch. During the labeling process, if the background details can be clearly observed through the clouds, we consider the clouds as "thin clouds"; otherwise, we consider them as "thick clouds." In addition, if more than half of the cloud pixels in an image are thin clouds, this image is considered as a thin-cloud image; otherwise, it is considered as a thick-cloud image.

Note that it is impracticable to manually annotate their accurate ground truth values. This is because the cloud reflectance and opacity are both in continuous values. In image matting, a current solution to this problem is to use synthetic data [20], [26]. We have followed this idea to generate a set of synthetic images with their "ground truth" labels for cloud reflectance maps, opacity maps, and attention maps. The thick-cloud images (where $\alpha(x) \approx 1$) and background images (with no clouds, where $\alpha(x) \approx 0$) in our training set are used to generate the synthetic images and their ground truth maps. The synthetic data generation process is shown in Fig. 3. We use the image regions that completely covered by thick clouds as the ground truth cloud reflectance of synthetic images. We use the images without any clouds as the ground truth background reflectance of the synthetic images. Then, the synthetic image can be generated by performing a linear combination of clouds and backgrounds based on (2), where a random opacity value is generated as the combination weights. To increase the diversity of the synthetic images, the background images and cloud reflectance maps are randomly rotated, flipped, and cropped. We select 38 background images and 34 thick-cloud images from the training set to synthesize 10 405 images for training the cloud matting networks. We select 23 background images and 43 thick clouds

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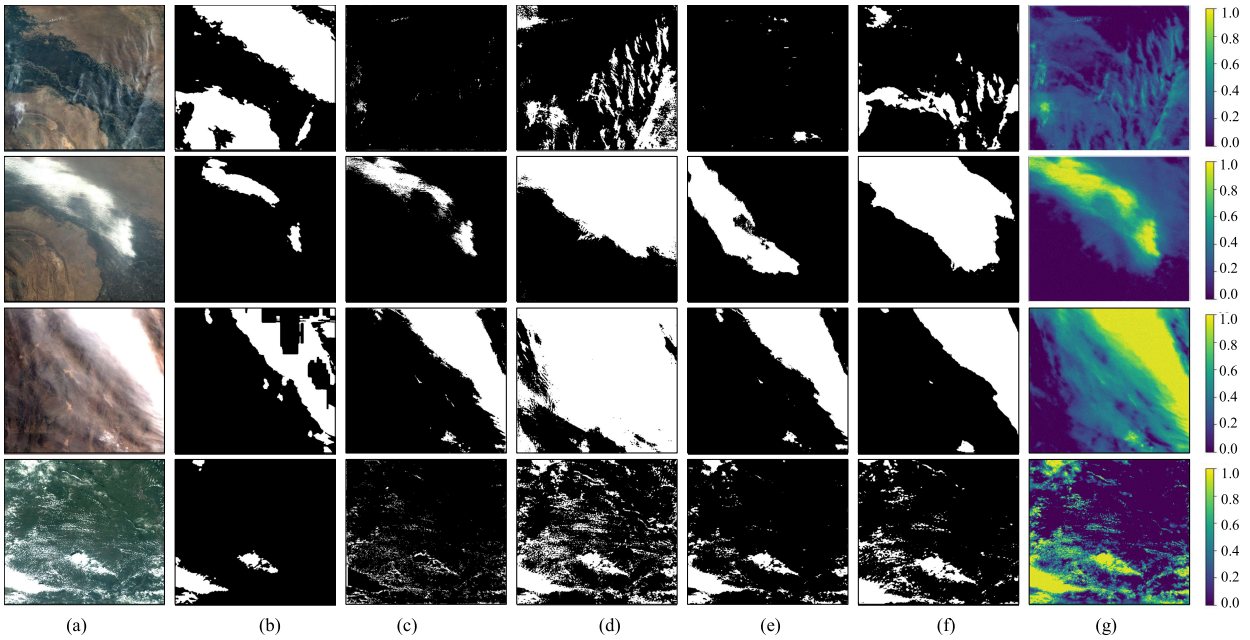



Fig. 4. (Better viewed in color) Some examples of cloud detection results of comparison methods. The first column shows the input cloud images. The second–sixth columns are the results of the comparison methods. The last column is the predicted cloud reflectance of our method (Cloud Matting + Proposed Networks). (a) Input images. (b) Progressive Refinement Scheme. (c) Scene learning. (d) FCN + CLS. (e) CloudFCN. (f) RS-Net. (g) Cloud matting.

from the testing set to synthesize 5934 images for evaluating cloud detection and cloud removal accuracy. Since we train our networks with synthetic images but test on real ones, we need more real images for the test phase to obtain more convincing results. Therefore, we leave more images in the testing set.

In addition to our GF-1 data set, we also test on two public cloud detection data sets: the GF1_Wuhan University (WHU) data set [5] and the Landsat-8 data set [56]. The GF1_WHU data set [5] consists of 108 images, and the Landsat-8 data set [56] consists of 96 images. Since we do not use images of the two data sets to train our model, we treat all images in these two data sets as the test ones. All images in the abovementioned two data sets are converted to 8-bit images and downsampled to $512 \times 512$ pixels. In addition, because the networks in our method are trained on RGB images, when testing on these two data sets, we have also selected these three bands accordingly.

## V. Experimental Results and Analysis

For the cloud detection task, the precision–recall (PR) curve and the "average precision (AP)" score are used as our evaluation metrics. The PR curve shows the relationship between the detection precision and the recall rate by changing different thresholds on a detection output

$$\text{Precision} = \frac{N_c}{N_c + N_f}, \quad \text{Recall} = \frac{N_c}{N_t} \tag{8}$$

where $N_t$ is the total number of cloud pixels in ground truth, $N_c$ is the number of correct detection of cloud pixels, and $N_f$ is the number of false alarm pixels.

For cloud reflectance and opacity prediction tasks, three different metrics are used, including the mean absolute error (MAE), mean squared error (MSE), and mean absolute percentage error (MAPE), which are defined as follows:

$$\begin{aligned} \text{MAE} &= \sum_i |y_i - \widehat{y}_i|/N \\ \text{MSE} &= \sum_i (y_i - \widehat{y}_i)^2/N \\ \text{MAPE} &= \sum_i |\frac{y_i - \widehat{y}_i}{\widehat{y}_i}|/N \end{aligned} \tag{9}$$

where $y$ and $\widehat{\mathbf{y}}$ are the predicted output and the ground truth reference, respectively. $i$ is the pixel id, and $N$ is the total number of pixels in an image.

### A. Detection Results

We compare our method with some recent cloud detection methods, including the progressive refinement [6], scene learning [8], fully convolutional networks + pixelwise classification (FCN+CLS) [14], CloudFCN [15], and RS-Net [16], on our test set. We also replace the encoder of the cloud matting network with VGG16 [57] and Resnet50 [46] to evaluate the performance of our framework. Fig. 4 shows some cloud detection examples. The first column shows the input cloud images. The second–sixth columns are the results of the comparison methods. The last column is the predicted cloud reflectance of our method.

Since there are great differences between thin clouds and thick clouds in visual appearance and detection difficulty, we evaluate our method based on the results of thin clouds and thick clouds separately. It can be seen from Table III and Fig. 5 that our method has higher cloud detection accuracy especially for those thin-cloud images, regardless of encoder's structure. For thick-cloud images, our method has comparable detection results on high recall areas with FCN+CLS [14],

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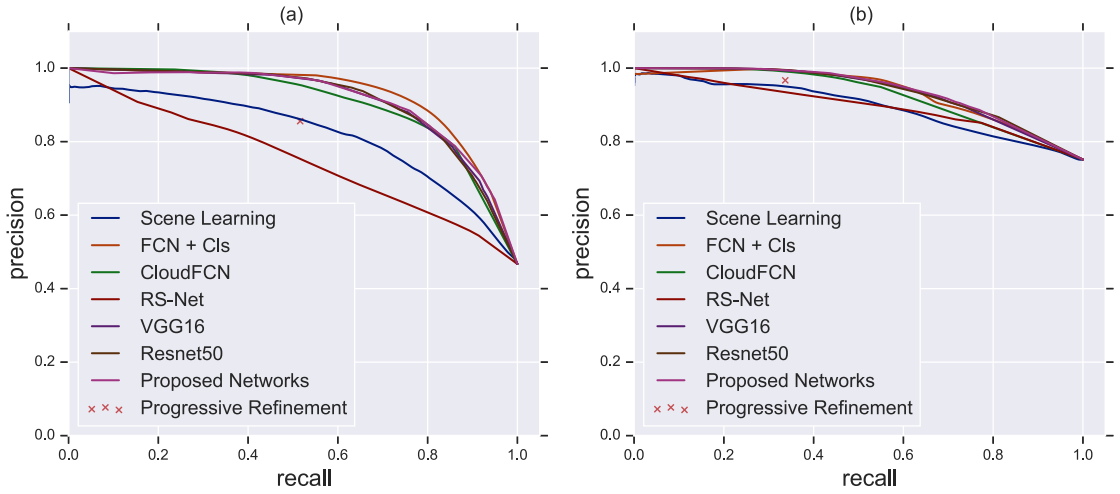



Fig. 5. (Better viewed in color) PR curves of different cloud detection methods. (a) Results on thick-cloud images. (b) Results on thin-cloud images.

TABLE III
COMPARISON OF CLOUD DETECTION RESULTS OF DIFFERENT METHODS. A HIGHER SCORE SUGGESTS A BETTER RESULT

| Method | $AP_{thick}$ | $AP_{thin}$ |
|---|---|---|
| Scene Learning [8] | 0.8200 | 0.8944 |
| FCN+Cls [14] | **0.9253** | 0.9332 |
| Progressive Refinement [6] | – | – |
| CloudFCN [15] | 0.9031 | 0.9261 |
| RS-Net [16] | 0.7518 | 0.8976 |
| Ours (Cloud Matting+VGG16) | 0.9110 | 0.9356 |
| Ours (Cloud Matting+Resnet50) | 0.9109 | 0.9370 |
| Ours (Cloud Matting+Proposed Networks) | 0.9139 | **0.9382** |

TABLE IV
COMPARISON RESULTS OF DIFFERENT METHODS ON THE GF1_WHU DATA SET [5] AND THE LANDSAT-8 DATA SET [56]. THE "AP" IS USED AS AN EVALUATION METRIC. A HIGHER SCORE SUGGESTS A BETTER RESULT

| Method | GF1_WHU [5] | Landsat-8 [56] |
|---|---|---|
| Scene Learning [8] | 0.4237 | 0.5282 |
| FCN+Cls [14] | 0.7594 | 0.7712 |
| Progressive Refinement [6] | – | – |
| CloudFCN [15] | 0.8105 | 0.6821 |
| RS-Net [16] | 0.5690 | 0.5724 |
| Ours (Cloud Matting+VGG16) | 0.7891 | 0.6896 |
| Ours (Cloud Matting+Resnet50) | 0.8035 | 0.6981 |
| Ours (Cloud Matting+Proposed Networks) | 0.7549 | 0.7074 |

and their curves are crossed with each other. As the PRS [6] only produces binary output masks, we cannot compute its AP in Table III and can only mark their precision and recall as a single point in Fig. 5 for comparison. The advantage of our method is not only suggested by the metrics we currently use but also in terms of the physical mechanism of the cloud images. Although our method is trained solely with synthetic data, the experimental result demonstrates that it still achieves comparable accuracy with other popular cloud detection methods on real data.

Table IV and Fig. 7 show the cloud detection results on the GF1_WHU data set [5] and Landsat-8 data set [56]. It can be seen that our method can not only obtain comparable AP scores with other cloud detection methods on the abovementioned two data sets but also extract the cloud reflectance accurately from the image. This suggests that our method can be applied to a variety of satellite platforms.

### B. Cloud Removal Evaluation

According to 3, once we have obtained the reflectance image and opacity of the cloud, the background can be easily recovered. In this way, we can evaluate the performance of cloud removal by calculating the MAE, MSE, and MAPE of the recovered images and their "ground truth." We compare

TABLE V
COMPARISON OF DIFFERENT METHODS ON THE CLOUD REMOVAL TASK. THE LOWER SCORES INDICATE A BETTER RESULT. THE BEST RESULTS IN EACH ENTRY ARE MARKED AS BOLD

| Method | MAE | MSE | MAPE |
|---|---|---|---|
| Homomorphic Filter [38] | 0.3400 | 0.1444 | 0.4312 |
| Deformed-Haze [39] | 0.0997 | 0.0169 | 0.2100 |
| Adaptive Removal [40] | 0.0655 | 0.0089 | 0.1449 |
| SM-DCP [41] | 0.1763 | 0.0485 | 0.3652 |
| Ours (Cloud Matting+VGG16) | 0.0599 | 0.0080 | 0.1180 |
| Ours (Cloud Matting+Resnet50) | 0.0648 | 0.0087 | 0.1306 |
| Ours (Cloud Matting+Proposed Networks) | **0.0570** | **0.0068** | **0.1140** |

our method with the four classical cloud removal methods: Homomorphic Filter [38], Deformed-Haze [39], Adaptive Removal [40], and sphere model improved dark channel prior (SM-DCP) [41]. We use our synthetic data sets to quantitatively assess the effects of cloud removal. From Table V, we can see our method achieves the best cloud removal result.

Fig. 6 shows some examples of cloud removal results for real images, where the column (a) shows the input image, column (b) shows the cloud removal result, column (c) shows the predicted opacity, and column (d) shows the predicted cloud reflectance. It can be seen that the thin cloud has been removed and the ground object has been nicely recovered.

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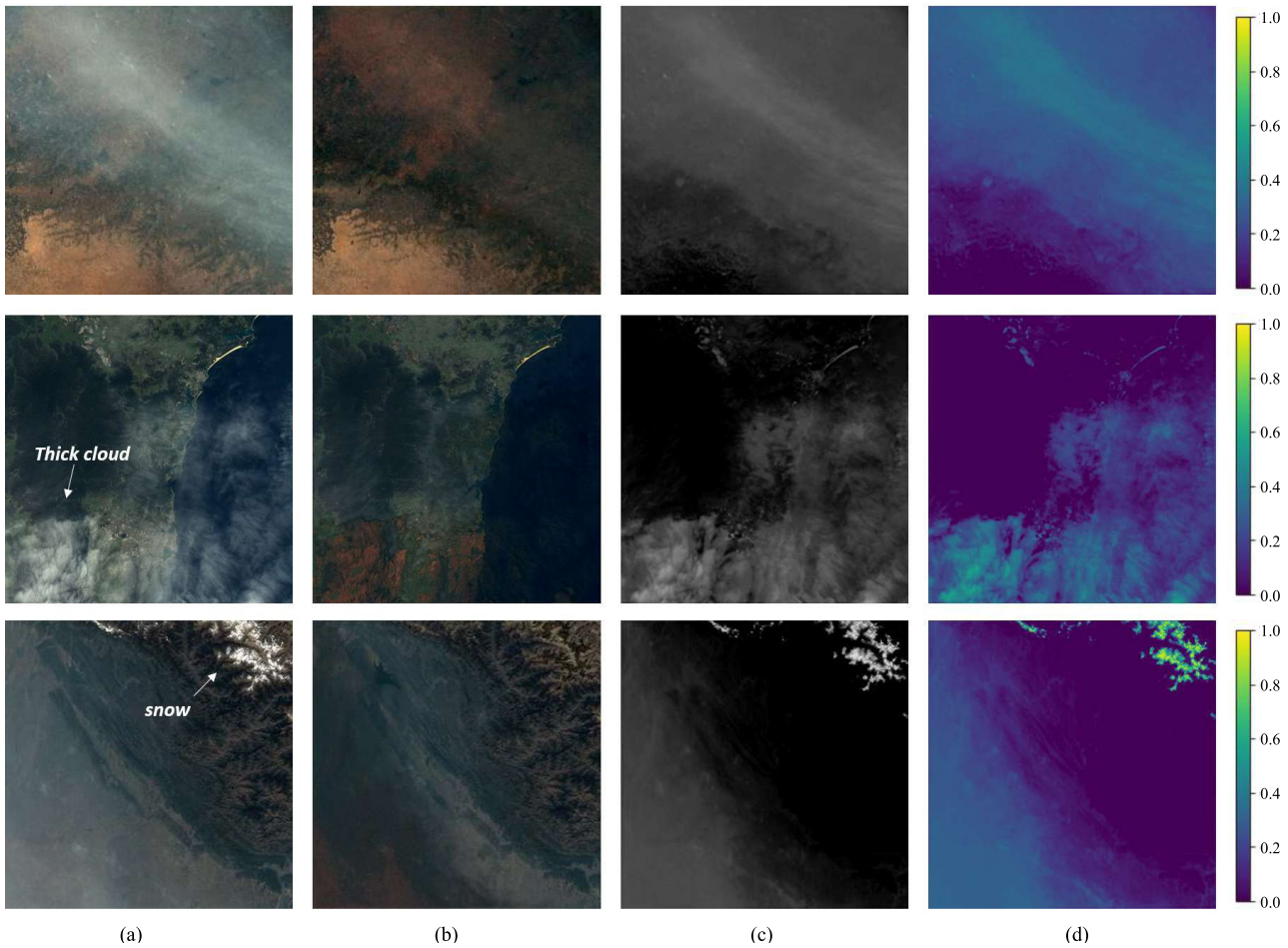



Fig. 6. (Better viewed in color) Some examples of the thin-cloud removal result of our method. (a) Input image. (b) Cloud removal result. (c) Predicted opacity (cloud cover assessment result). (d) Predicted cloud reflectance. The last row shows a failure case of our method; there is no cloud in this image, but our model has wrongly predicted the snow as the cloud. This may be because our training set does not contain any snow images.

TABLE VI

ABLATION STUDIES OF CLOUD REFLECTANCE AND OPACITY PREDICTION. ABLATIONS ARE PERFORMED ON: 1) ATTENTION MECHANISM; 2) FF; AND 3) BN

| **Ablations** | | | **Cloud Reflectance Prediction** | | | **Cloud Opacity Prediction** | | |
|---|---|---|---|---|---|---|---|---|
| Attention | Feature Fusion | Batch-Norm | $\varepsilon_{MAE}$ | $\varepsilon_{MSE}$ | $\varepsilon_{MAPE}$ | $\varepsilon_{MAE}$ | $\varepsilon_{MSE}$ | $\varepsilon_{MAPE}$ |
| × | × | × | 7.44% | 2.70% | 86.3% | 9.86% | 4.47% | 79.4% |
| × | × | ✓ | 7.33% | 2.28% | 85.0% | 9.84% | 3.83% | 78.23% |
| × | ✓ | ✓ | 4.90% | 0.717% | 72.5% | 9.55% | 2.25% | **68.3%** |
| ✓ | ✓ | ✓ | **3.04%** | **0.302%** | **71.6%** | **6.56%** | **2.10%** | 73.7% |

In the fourth row of this figure, as the cloud opacity is close to 1, the background under the thick-cloud region is hard to recover, thus leading to slight color distortion. The last row of this figure shows a failure case of our method, where there is actually no cloud in this image but our model has wrongly identified the snow as the cloud. This is mainly because our training set does not contain any snow images, and this can be easily improved by adding more snow images for training. Since this is not the focus of this article, we do not make any further evaluations on distinguishing cloud and snow pixels.

### *C. Ablation Analyses*

In this experiment, the ablation analyses are made to analyze the importance of each technical component of our method, including: 1) attention mechanism; 2) FF; and 3) BN. The baseline methods are first evaluated, and then, we gradually integrate these techniques. Table VI shows their comparison results of the reflectance and opacity on the subtesting set. Fig. 8 shows their comparison of the cloud detection results on the whole testing set. The integration of the first two techniques yields a noticeable improvement of the prediction accuracy of cloud reflectance and opacity, while the BN is trivial for the improvement of accuracy. Nevertheless, the model trained with BN still benefits from a faster convergence speed. The reason behind the improvement is twofold: on the one hand, as there is a strong correlation between the cloud reflectance and opacity, the attention map helps to eliminate the coupling between these two tasks; on the other hand, FF is

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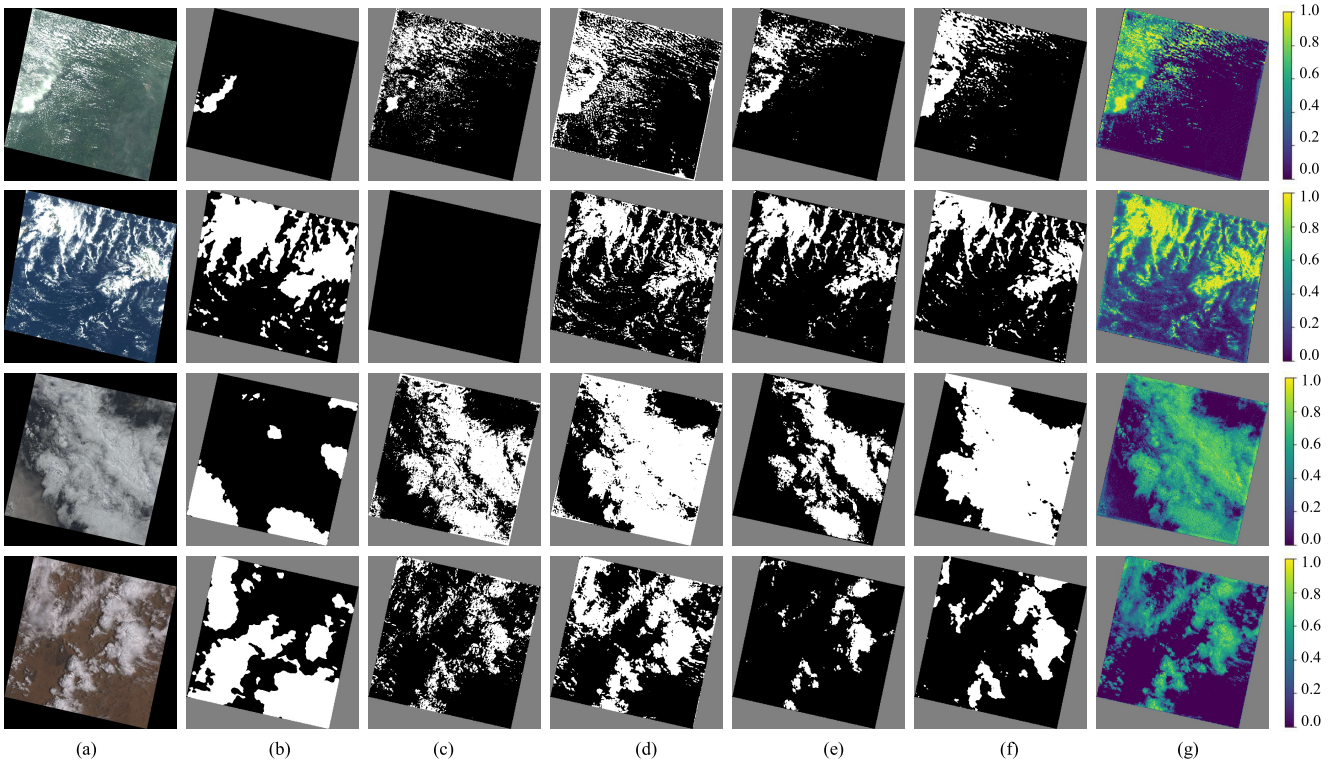



Fig. 7. (Better viewed in color) Some examples of the cloud detection results of comparison methods on the GF1_WHU data set [5] (the first two rows) and Landsat-8 data set [56] (the last two rows). The first column shows the input cloud images. The second–sixth columns are the results of the comparison methods. The last column is the predicted cloud reflectance of our method (Cloud Matting + Proposed Networks). The parts marked in gray correspond to the "black areas" in the input image, and the parts marked in black and white correspond to the background and cloud separately. (a) Input images. (b) PRS. (c) Scene learning. (d) FCN + CLS. (e) CloudFCN. (f) RS-Net. (g) Ours.

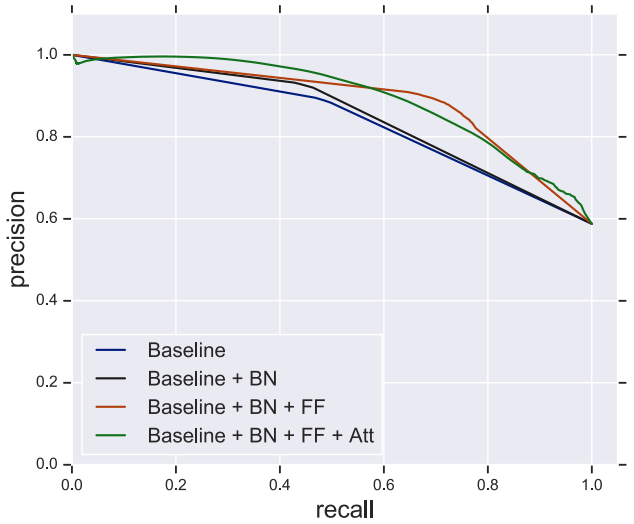



Fig. 8. (Better viewed in color) Ablation studies of cloud detection accuracy on testing data. Ablations are performed on: 1) attention mechanism (Att); 2) FF; and 3) BN.

beneficial for predicting a more detailed output for some areas, such as some small pieces of clouds and the clouds with sharp edges. Fig. 9 shows the effectiveness of the attention mechanism. We can see that in the area covered by clouds, the values of its attention map are larger, no matter where it is covered by thin clouds or thick clouds. Therefore, the attention map can be used to guide the training of the network and make the networks concentrate on the area covered by clouds and obtain better cloud detection results. Meanwhile, predicted values without attention are smaller than those with attention.

### D. Cloud Image Montage

The proposed framework can also be used for another important application: cloud image montage, i.e., to transplant the cloud in one image to another background image. This can be simply implemented by the following transformation:

$$I'(x) = R_c^B(x) + [1 - \alpha^B(x)]I^A(x) \tag{10}$$

where $I^A(x)$ is an image from a background image set A, $R_c^B(x)$ and $\alpha^B(x)$ are the predicted cloud reflectance and opacity of an image from a cloud image set B, and $I'(x)$ is the generated cloud montage output. Fig. 10 shows some examples of our cloud montage generation results. In Fig. 10, some very high-resolution aerial images from Google Earth are used as the background images, and the clouds from GF-1 images from our test set are used as the foreground cloud styles.

The abovementioned process can be considered as a new way of data augmentation, which helps us to generate some hard examples. It may have great potential for improving the performance of some RS applications, such as occluded target detection, scene recognition, and image segmentation.

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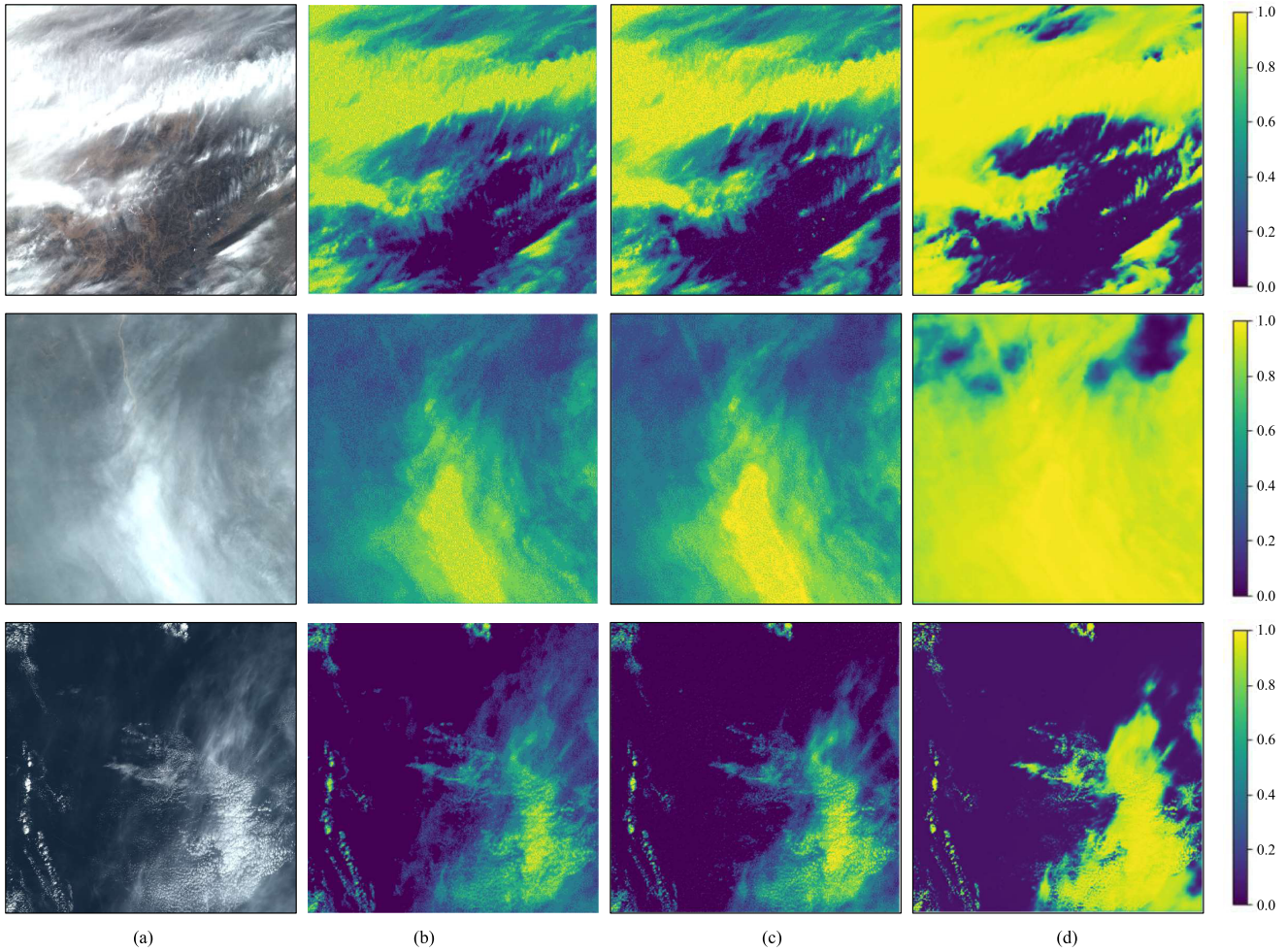



Fig. 9. (Better viewed in color) Effectiveness of the proposed attention mechanism. The first column shows the input images. The second and third columns show the predicted reflectance maps w/o and w/ the help of attention loss. The last column shows the attention maps. (a) Input Image. (b) Reflectance without attention. (c) Reflectance with attention. (d) Attention map.

TABLE VII
COMPARISON ON MODEL PARAMETERS, FLOPS, AND INFERENCE TIME

| Method | **Params** | **FLOPs** | **Inference time** |
|---|---|---|---|
| FCN+Cls [14] | 18.5M | 12.4G | 0.103s |
| CloudFCN [15] | 23.3M | 155M | 0.054s |
| RS-Net [16] | 7.9M | 39M | 0.288s |
| Ours (Cloud Matting+VGG16) | 74.0M | 25.8G | 0.224s |
| Ours (Cloud Matting+Resnet50) | 139M | 17.3G | 0.410s |
| Ours (Cloud Matting+Proposed Networks) | 79.5M | 22.1G | 0.210s |

### *E. Computational Complexity, Parameters, and Speed*

We use three different metrics to compare the computational complexity, parameters, and speed of our method with other CNN-based cloud detection methods. In Table VII, we record the number of model parameters (Params), the number of floating-point operations (FLOPs), and the inference time of different models. We use images of $512 \times 512$ pixels to compute FLOPs and inference time, and test on an Nvidia GeForce RTX 2080 Ti graphics card. Compared with other methods, our method has more parameters but has comparable inference time with RS-Net [16]. This is because our method uses different branches to predict the reflectance, the opacity, and the attention map, which requires more parameters and memories.

## VI. DISCUSSION

Although the experimental results on the three satellite image data sets demonstrate the effectiveness of our method, it still has some limitations.

1) We do not take the shadow detection into consideration. Since shadows often appear with clouds, it is also important to consider the shadow detection in our framework. In fact, the proposed model formulated in 2 can be simplified to $I(x) = [1 - \alpha(x)]R_b(x)$ by setting $R_c(x) = 0$. Then, our framework can be naturally extended to shadow detection tasks.
2) As illustrated in the last row of Fig. 6, our method may produce a wrong detection result when there is snow in the image. The reason behind this could be the limited snow samples in our training set. To improve the detection performance on snow-covered regions, we may simply add more snow images to our training set.

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Fig. 10. Some examples of the cloud image montage results. The first row shows the input images from image set A. The second row shows the input images from image set B. The third row shows the generation outputs by combining the cloud style of the image set A and backgrounds of the image set B.

3) We use a simple data synthesis method to support the training of our model. Despite our primary verification and the promising results obtained, there is still much room for improvement in our method, especially for data synthesis. In our future research, we may design a more sophisticated data synthesis process (e.g., nonuniform opacity by using adversarial training, where we have already obtained some promising results [58]) to generate more realistic cloud images.

## VII. Conclusion

We propose a brand new method for cloud detection in RS images, which inherently incorporates the cloud imaging mechanism and jointly deals with three different but related problems, such as cloud detection, cloud cover assessment, and cloud removal, under the same framework. Different from previous methods that consider cloud detection as a pixelwise binary classification problem, we reformulate cloud detection as a mixed energy separation problem. The experimental results on the three satellite image data sets demonstrate the effectiveness of our method. In addition, the proposed framework can also be used to synthesize cloud images of specific style, which can be considered as a new way of data augmentation and may have great potential for improving the performance of many RS applications, such as occluded object detection and recognition.

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